🤌 XAI: Quantifying Data Value and Influence in Machine Learning

FAMAF - UNC

First Semester 2026

eXplainable
Artificial Intelligence

🧉 Disclaimer ------

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

Quantifying Data Value and Influence in Machine Learning

Combining insights from:

What is Your Data Worth? Equitable Valuation of Data (Ghorbani & Zou, ICML 2019)

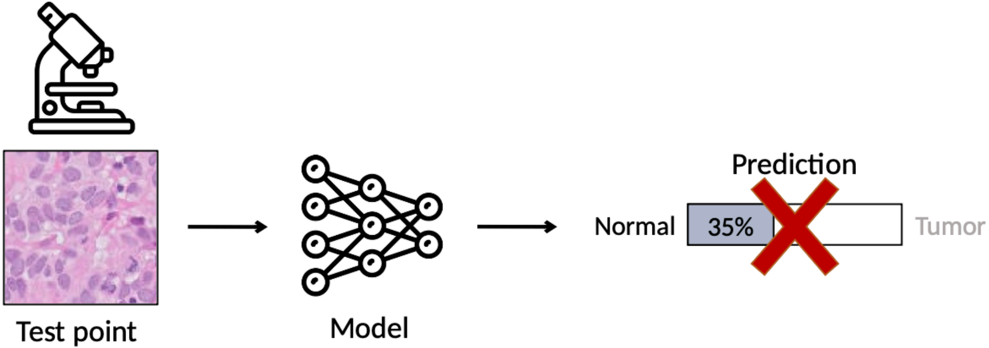
Understanding black-box predictions via influence functions (Koh & Liang, ICML 2017)

🧉 The Data Valuation & Influence Problem ------

If data is fuel, we need a principled way to measure its value and its effect on our models. Stakeholders have different priorities:

- **ML Engineers:** Assess heterogeneous sources, debug model behavior, and ensure data quality.
- **Data Vendors:** Determine fair pricing for buying and selling data.
- **Individuals/Data Producers:** Understand compensation or credit for contributed data.

🧉 The Baseline: Leave One Out Method ------



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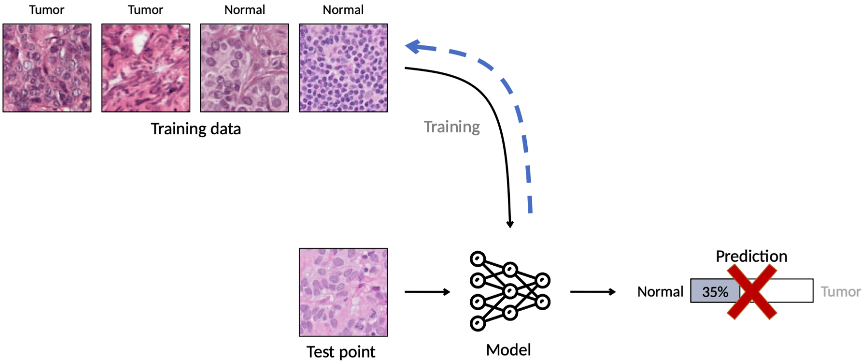

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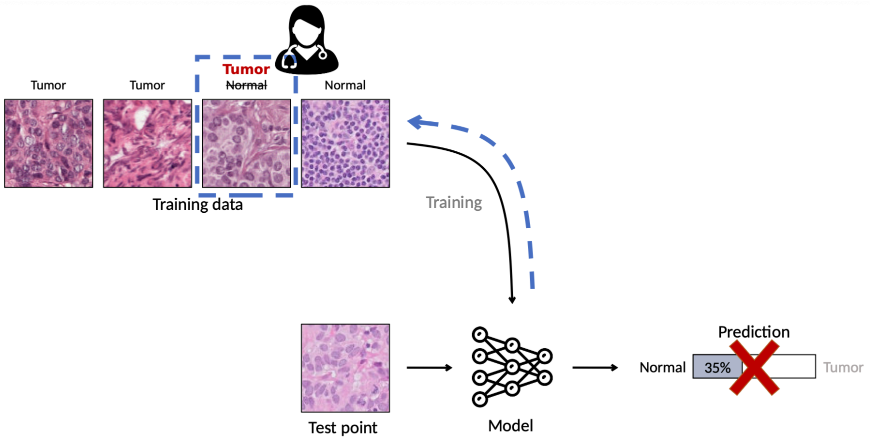

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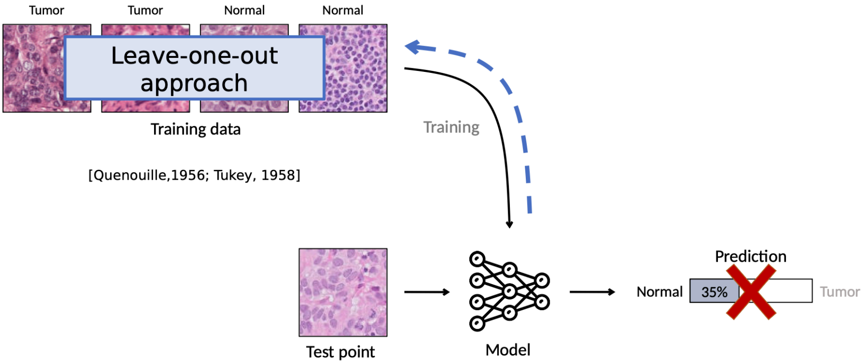

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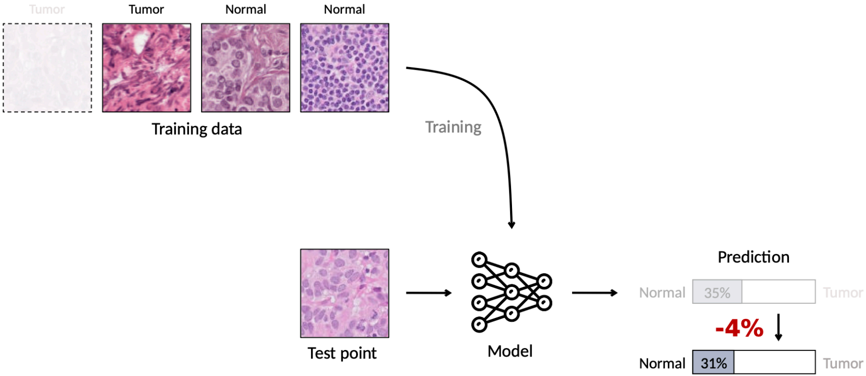

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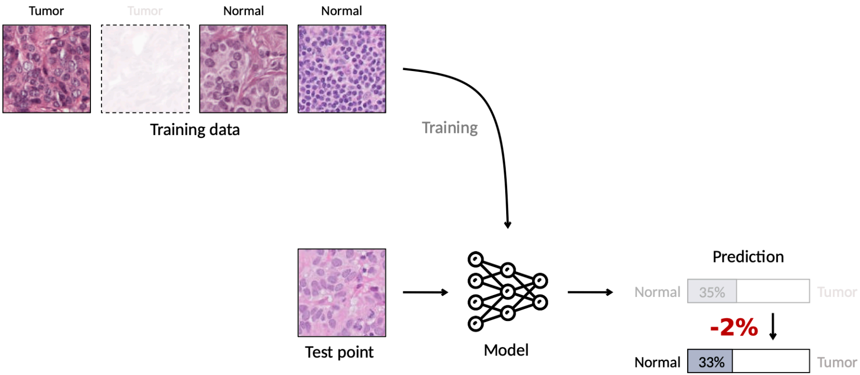

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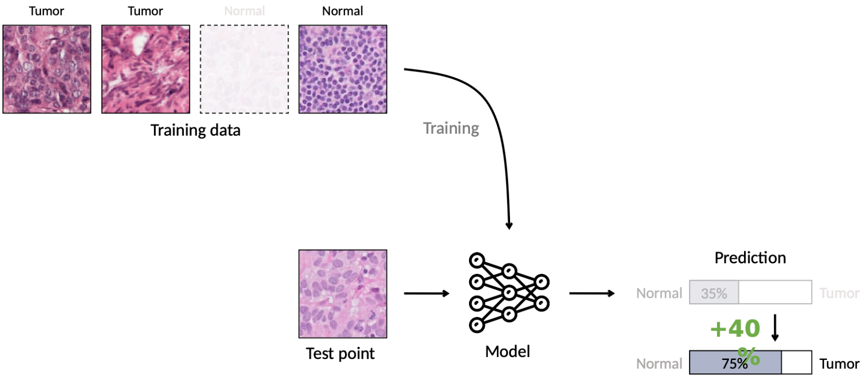

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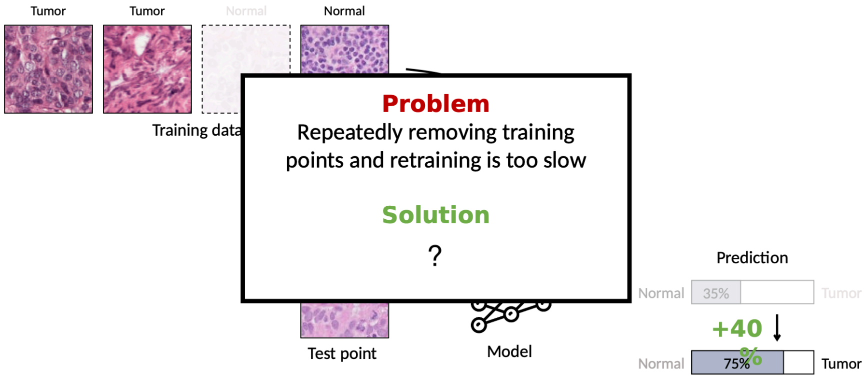


Problem

Repeatedly removing training points and retraining is too slow

Solution

?

🧉 The Bottleneck ------

Leave-One-Out (LOO) is conceptually simple but practically flawed:

1. **Computationally intractable:** Retraining the model for every single data point is impossible for large datasets.
2. **Ignores interactions:** It doesn't account for how data points interact with one another in subsets.

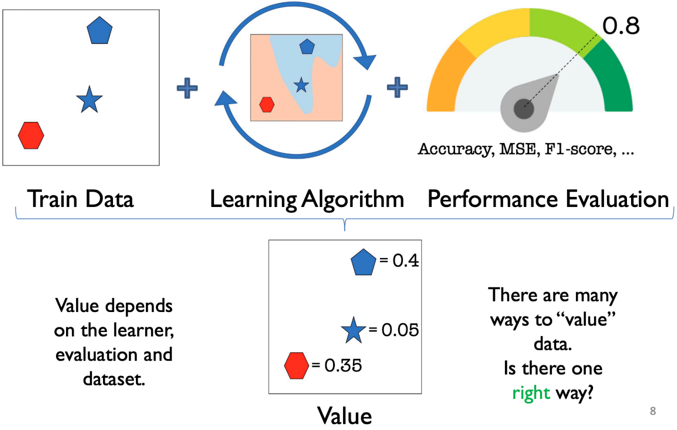
Two frameworks address these issues: **Data Shapley** and **Influence Functions**.

Approach 1: Data Shapley

Equitable Valuation of Data

(Ghorbani and Zou 2019)

🧉 Ingredients of Data Value ------



Value depends on the learner, evaluation and dataset.

There are many ways to "value" data.
Is there one right way?

8

🧉 Desirable Properties of Valuation ------

A fair valuation method must satisfy three fundamental axioms from cooperative game theory:

1. **Null Element:** If adding a point to *any* subset of training data never changes the model's performance, its value is exactly 0.
2. **Symmetry:** If adding point A or point B to any subset always results in the exact same performance change, they must receive the same value ($Value(A) = Value(B)$).
3. **Linearity:** If the performance metric is a sum of performances on individual tasks, the data value must reflect the sum of values for those individual tasks.

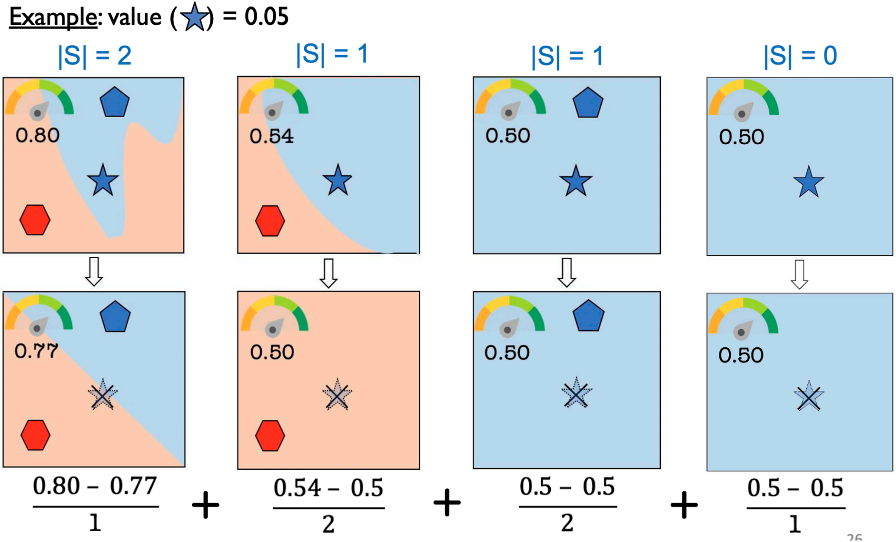
🧉 The Data Shapley Value ------

The only data value mathematically guaranteed to satisfy these three properties is based on marginal contributions across all possible subset sizes.

$$Value(data\ k) = \sum_{S \subseteq D\ \{k\}} \frac{performance(S \cup \{k\}) - performance(S)}{\binom{n-1}{|S|}}$$

- Evaluates the expected contribution to all possible sizes of train data samples.
- Directly applies Lloyd Shapley's cooperative game theory to individual ML data points.

🧉 The Data Shapley Value Example ------



🧉 Efficient Approximation: TMC-Shapley ------

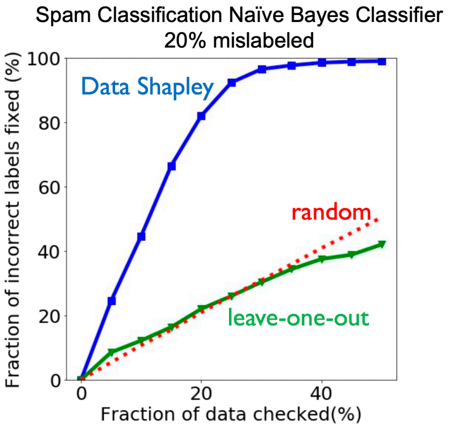
Exact calculation is computationally infeasible for realistic datasets because it grows exponentially.

Truncated Monte-Carlo (TMC) Approximation:

1. Sample a random permutation of the training data.
2. Add one point at a time based on the sampled permutation.
3. Re-train the model to capture the marginal increase in performance.
4. **Truncate** the calculation when performance saturates to save computational resources.

🧉 Application 1: Identifying Low-Quality Data ------

Result: Ordering points from most negative Shapley value to positive allows for rapid dataset cleaning. Examining just ∼30% of the data uncovers nearly all mislabeled examples.



🧉 Application 2: Identifying Essential Data ------

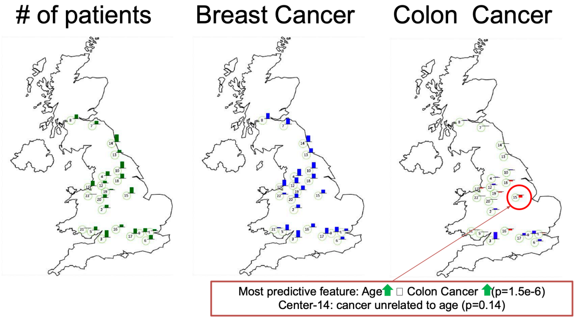
Case Study: UK Biobank

- Dataset spanning 500,000 individuals across 22 centers in the UK.
- Centers were evaluated as singular "data sources" for predicting Breast and Colon Cancer.

🧉 Application 2: Identifying Essential Data ------

Insights from Shapley:

- For Colon Cancer, a massive center received a *negative* Shapley value.
- Investigation revealed the model heavily relied on *Age* as a predictive feature, but the specific anomalous center exhibited colon cancer rates entirely independent of age.



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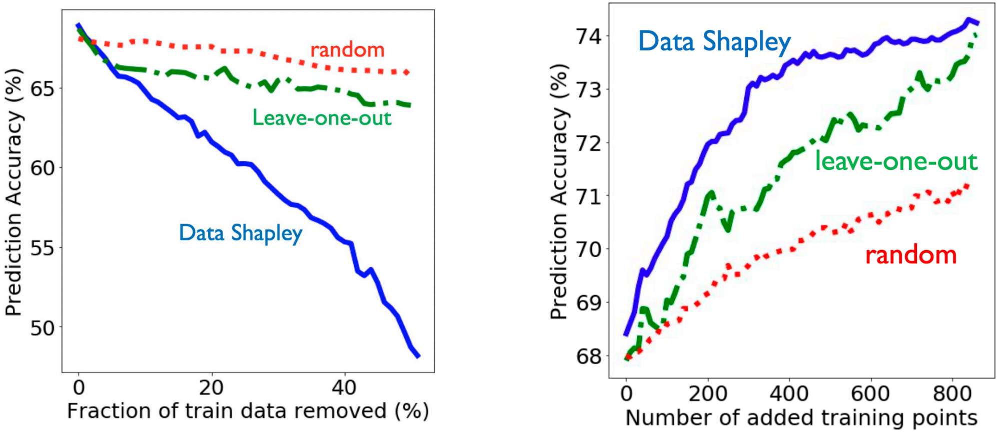


Application 3: Domain Adaptation ------

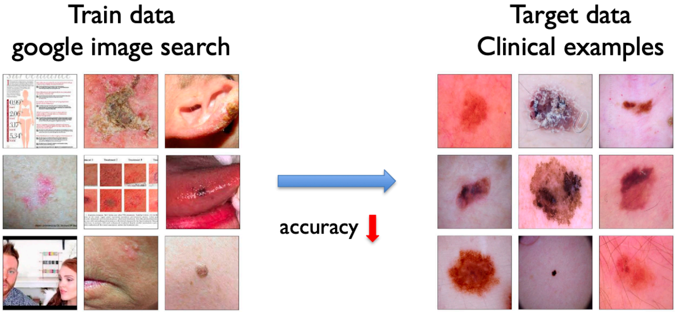
What if the training data differs in quality, distribution, or class balance from the test environment?

Shapley Adaptation Strategy:

1. Compute Shapley values relative to the target test distribution.
2. Remove data with negative values.
3. Reweight the remaining training data based on relative positive weights.

🧉 Domain Adaptation Results (Skin Lesion) ------

Training on noisy Google Image Search data to test on the clean, clinical HAM10000 dataset. Applying Shapley weights improved classification by $\approx 25\%$.



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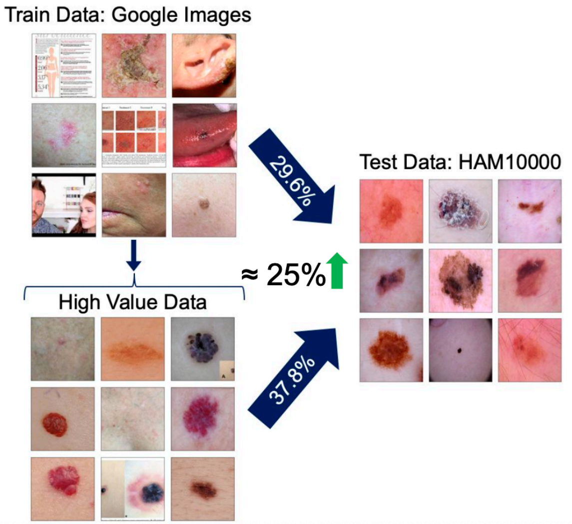
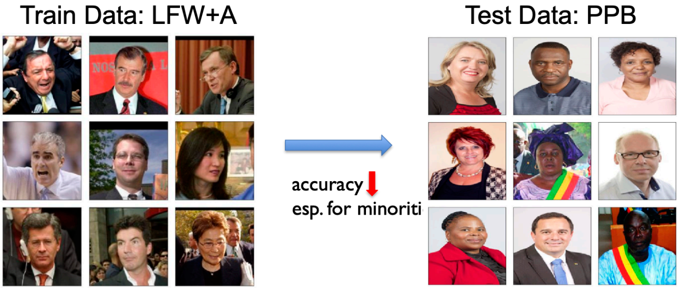


🧉 Domain Adaptation Results (Gender Detection) ------

Models trained on LFW+A (heavily biased toward white males) fail on balanced datasets like PPB. Shapley values easily identify out-of-distribution elements; adaptation increases accuracy on underrepresented minorities by $\approx 7\%$.



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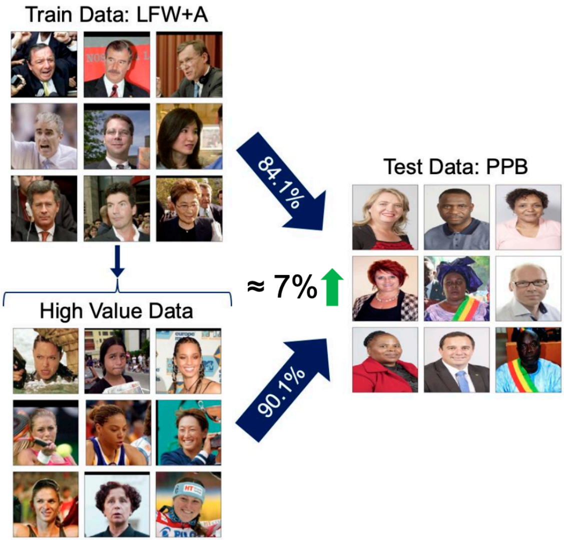


Approach 2: Influence Functions

Understanding black-box predictions via influence functions

(Koh and Liang 2017)

🧉 The Influence Function Approximation ------

$$\overbrace{\ell\left(z_{test}, \hat{\theta}_{-z_{train}}\right) - \ell\left(z_{test}, \hat{\theta}\right)}^{\text{Change in loss on after removing}} \approx \overbrace{\nabla_{\theta} \ell\left(z_{test}, \hat{\theta}\right)^{T}}^{\text{Model's representation of}} \overset{\text{Effect of other training points}}{H_{\hat{\theta}}^{-1}} \overbrace{\nabla_{\theta} \ell\left(z_{train}, \hat{\theta}\right)}^{\text{Model's representation of}}$$

Gradient of loss on

Gradient of loss on

Inverse of the Hessian

Doesn't require retraining!

🧉 Interpretation: Model Representations ------

$$\ell(z_{\text{test}}, \hat{\theta}_{-z_{\text{train}}}) - \ell(z_{\text{test}}, \hat{\theta}) \approx \nabla_\theta \ell(z_{\text{test}}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_\theta \ell(z_{\text{train}}, \hat{\theta})$$

Gradient = model's representation

$\nabla_\theta \ell(z_{\text{test}}, \hat{\theta})$ encodes how the model "sees" z_{test}

$H_{\hat{\theta}}^{-1}$ = effect of other training points

Accounts for the curvature induced by the full training set

- High influence ⇒ test and train have **similar model representations**
- If the resulting value is **positive** ⇒ the training point was **harmful** to that prediction
- If the resulting value is **negative** ⇒ it was **helpful**

🧉 Removing Single Points: Validation ------

Logistic regression (MNIST): each point = removing one training example. Influence function estimate tracks actual LOO change closely.

🧉 Limitations ------

Key assumptions required

- What assumptions on models are needed to calculate influence functions?
- Can you calculate influence functions for a neural network?

$$\hat{\theta}_{\epsilon,z} \stackrel{\text{def}}{=} \arg\min_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^{n} L(z_i,\theta) + \epsilon L(z,\theta)$$

- **Non-convexity:** derivation assumes a convex loss landscape
- **Non-convergence:** derivation assumes the optimizer has converged

"If Influence Functions are the Answer, Then What is the Question?"

Bae, Ng, Lo, Ghassemi & Grosse

Summary ------

Both approaches aim to link model behavior back to the training data.

Data Shapley:

- Axiom-backed framework moving beyond LOO by measuring expected contributions over all subsets.
- Excellent for data valuation, pricing, cleaning, and domain adaptation.
- Computationally intensive (requires TMC approximation).

Influence Functions:

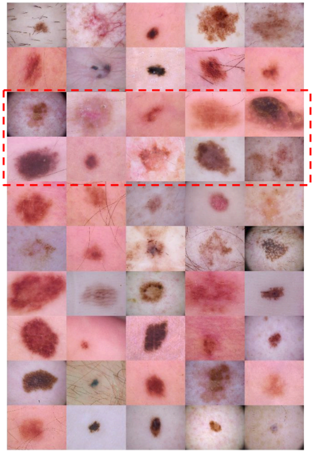
- Efficient gradient-based calculation—no retraining needed.
- Excellent for fast debugging, fairness analysis, and understanding specific predictions.
- Limited by strict mathematical assumptions (convexity, convergence) in complex models.

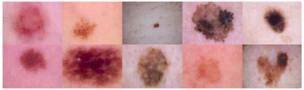
Extra: Distributional Shapley

A More Intrinsic Measure of Value

(Ghorbani et al. 2020)

🧉 Extra! (Distributional Shapley) ------





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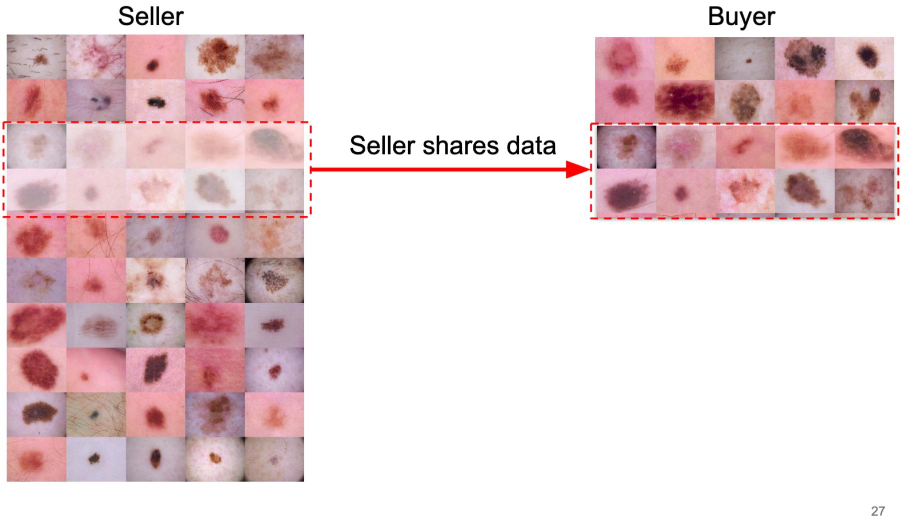

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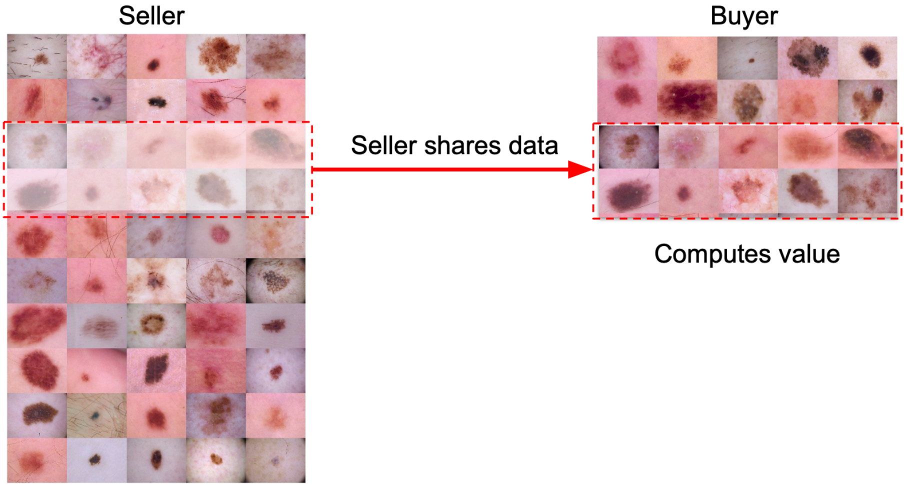

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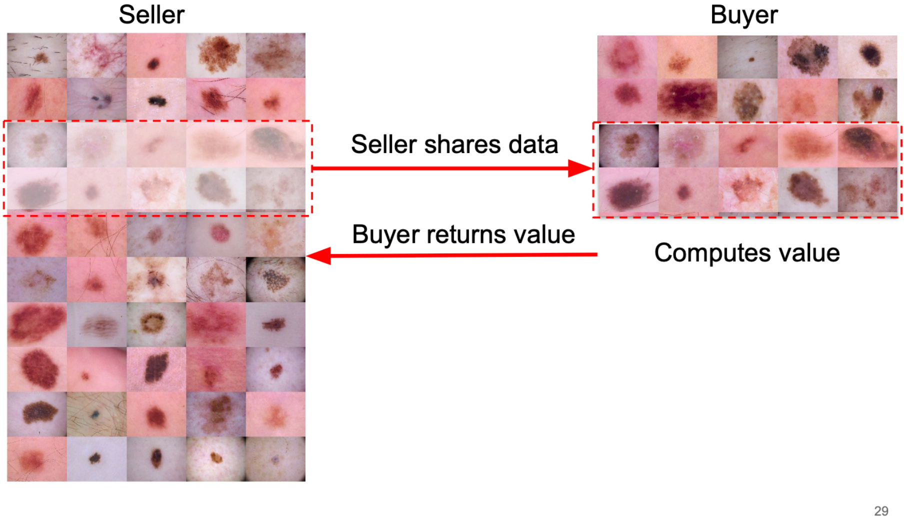

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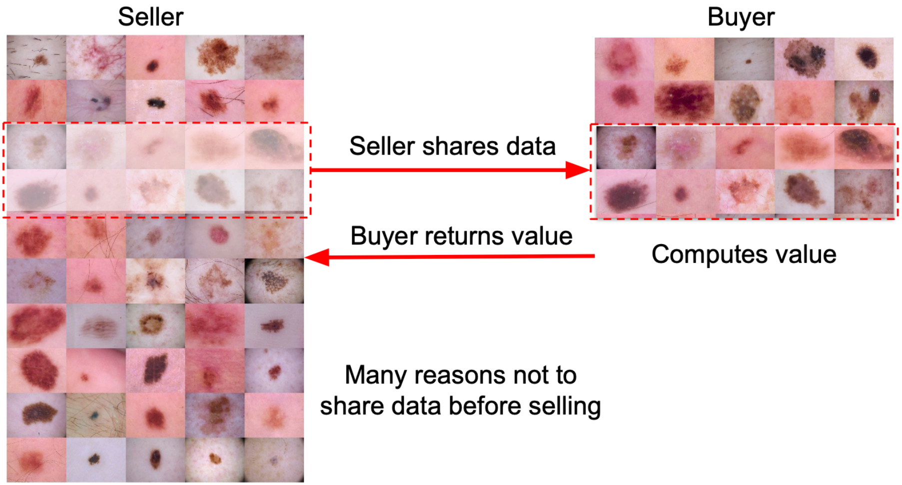

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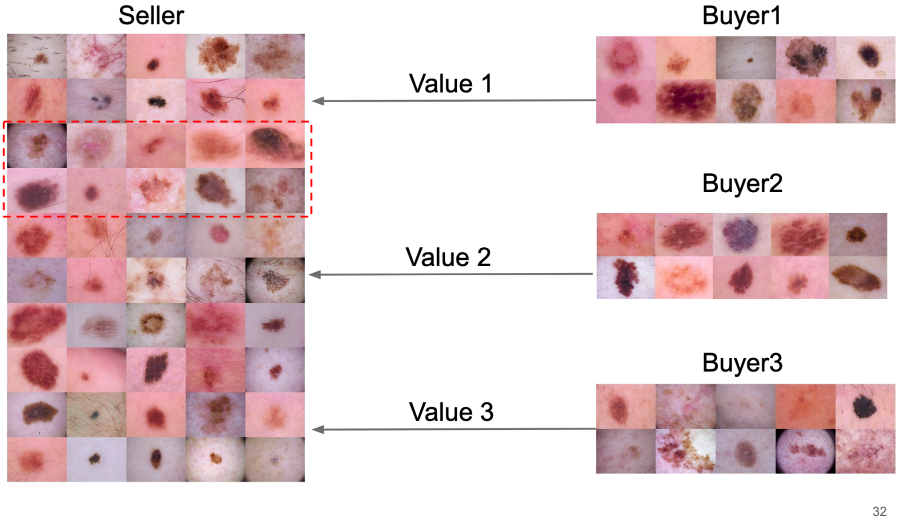

Thank You!

References ------ |

Ghorbani, Amirata, Michael P. Kim, and James Zou. 2020. "A Distributional Framework for Data Valuation." *Proceedings of the 37th International Conference on Machine Learning*, ICML'20.

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Koh, Pang Wei, and Percy Liang. 2017. "Understanding Black-Box Predictions via Influence Functions." *International Conference on Machine Learning*, 1885–94.